



Influence of wall coverings of 3D-printed vocal tract models on measured transfer functions

Peter Birkholz¹, Dominik Schäfer¹, Patrick Häsner¹, Jihyeon Yun¹, Iris Kruppke², Rémi Blandin³

¹Institute of Acoustics and Speech Communication, TU Dresden, Germany

²Institute of Textile Machinery and High Performance Material Technology, TU Dresden, Germany

³CNRS, IETR-UMR 6164, Université de Rennes, France

peter.birkholz@tu-dresden.de

Abstract

Physical replicas of the vocal tract are often produced by 3D-printing. A proven method to obtain the acoustic transfer function of these models is based on the principle of reciprocity, where an external sound source is used and the acoustic response is measured inside the model at the glottal end. However, earlier investigations showed that for some model geometries the measured transfer functions deviated from the expected resonance pattern due to spurious poles and zeros. These artifacts are probably caused by sound transmission through the model walls, or by whole-body vibrations of the replicas. In this empirical study, two methods to reduce these artifacts were compared, namely wrapping the models with sound-absorbing fabric and embedding the models in sand. Axisymmetric 3D-printed tubes for ten different vowels were analyzed. The results showed that both methods reduce artifacts and at the same time improve the repeatability of the measurements.

Index Terms: vocal tract replicas

$P_1(\omega)/P_{\text{ref}}(\omega)$. In addition, the use of a sine sweep as the excitation signal also enables the determination of the *linear* vocal tract response, which is free of the harmonic distortions generated by the loudspeaker and microphone [11, 6].

Despite these positive aspects of the measurement method, the transfer functions obtained for some 3D-printed vocal tract models may still show unexpected artifacts. For example, Birkholz et al. [12] used this method to measure the transfer functions of 3D-printed axisymmetric tube models for different vowels. The models were printed with the hard plastic material PLA (polylactic acid) with an infill ratio of 100% and a wall thickness of 3 mm. For some of the models, the transfer function deviated from the expected pattern of an all-pole transfer function below 5 kHz and showed traces of spurious poles and zeros between 1–3 kHz. This may prevent the accurate determination of the frequencies and bandwidths of some of the resonances.

1. Introduction

The increasing use of 3D printers has led to many studies in which 3D-printed models of the vocal tract are used to investigate speech and singing. These models range from simple axisymmetric tube models [1, 2, 3, 4] to geometrically realistic models based on MRI scans of the vocal tract [5, 6, 7, 8, 9]. For their acoustic characterization, many studies determine the volume velocity transfer function $H(\omega) = U_m(\omega)/U_g(\omega)$, where $U_g(\omega)$ is the volume velocity at the glottis and $U_m(\omega)$ is the volume velocity at the mouth opening.

A widely used method to determine $H(\omega)$ is based on the principle of reciprocity [5, 6, 10]. In this method, a loudspeaker positioned in front of the model emits a broadband sweep signal $P_s(\omega)$ towards the mouth of the model to excite the vocal tract resonances. At the same time, a microphone measures the sweep response $P_1(\omega)$ inside the model at the glottal end. Kitamura et al. [5] showed that $P_1(\omega)$ and $H(\omega)$ are related according to

$$P_1(\omega) = (1 + Z_{\text{rad}}(\omega)/Z_0) \cdot P_s(\omega) \cdot H(\omega), \quad (1)$$

where $Z_{\text{rad}}(\omega)$ is the radiation impedance of the model, and Z_0 is the characteristic impedance of a plane wave. Kitamura et al. [5] and Delvaux and Howard [6] approximated $H(\omega)$ as $P_1(\omega)/P_s(\omega)$, where $P_s(\omega)$ was measured as the free-field frequency response of the loudspeaker (without a vocal tract model). Fleischer et al. [10] showed that a more accurate approximation of $H(\omega)$ is obtained when $P_1(\omega)$ is divided by a reference pressure $P_{\text{ref}}(\omega)$ measured directly in front of the closed mouth of the vocal tract model, i.e., $H(\omega) \approx$

The cause for these artifacts is not entirely clear. It is unlikely that they are caused by higher-order modes, as these modes should only occur at higher frequencies in axisymmetric tube models [13]. However, the artifacts might originate from the transmission of sound through the walls of the models, or from whole-body vibrations of the models. There are several options to reduce these effects. Fleischer et al. [10] coated their models with a 1 cm thick layer of plaster to increase the mass of the walls and prevent sound transmission through the walls. This effectively avoided the artifacts, but the coating with plaster is very time-consuming and cannot be easily reversed. In contrast, Birkholz et al. [8] wrapped their models in multiple layers of sound-absorbing fabric to reduce sound transmission through the vocal tract walls. Finally, Blandin et al. [14] embedded their models in sand to absorb the mechanical vibrations of the model walls.

While all these measures seem to improve the quality of the measured transfer functions to different degrees, they have not been systematically compared for 3D-printed vocal tract models of different shapes. Therefore, the goal of the present study was to compare how the methods of wrapping the models with sound-absorbing fabric and embedding the models in sand reduce artifacts and improve the repeatability of measurements compared to using the models without any wall covering. Compared to coating with plaster, both of the methods used here are easier to apply. The present study extends a recent parallel study [15], which used only three instead of ten different model geometries, but supported the analysis with 3D-acoustic simulations.

2. Method

2.1. Vocal tract models

The vocal tract models used here were previously introduced in [12] and are available as 3D-printable STL files in the supplemental material of that publication (<https://www.vocaltractlab.de/index.php?page=birkholz-supplements>). There are ten models in total, all designed as straight tubes with circular cross-sections and different area functions. Nine models represent the area functions of the German vowels /a, e, i, o, u, ε, ø, y, ə/ of a male speaker, which were exported from the articulatory speech synthesizer VocalTractLab 2.3 [16] (www.vocaltractlab.de). The tenth model is a uniform cylinder with a length of 17 cm and a diameter of 1.7 cm. All models have a wall thickness of 3 mm and two flanges with diameters of 57 mm at both ends. All ten models were 3D-printed on an Ultimaker 3 printer using the stiff plastic material PLA (polylactic acid) with an infill ratio of 100 %. As examples, the three models for /a, e, i/ are shown in Figure 1a.

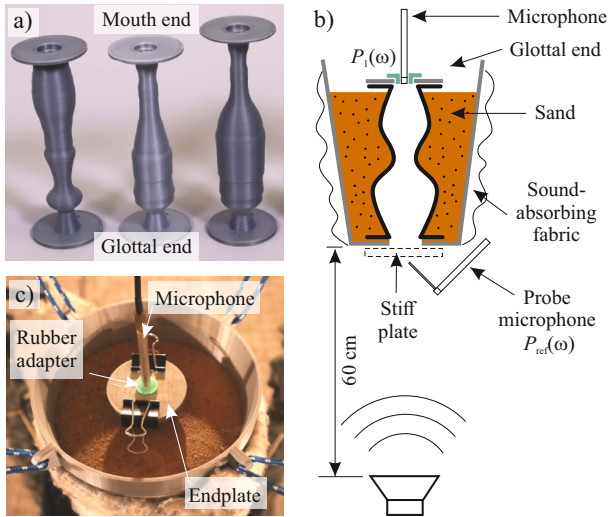


Figure 1: a) Tube models for the vowels /a,e,i/ (from left to right). b) Measurement setup with the tube model embedded in sand. c) View from above into the sand bucket. The glottal end of the tube with the microphone attached sticks out of the sand.

2.2. Measurements

The measurement setup is shown in Figure 1b and is similar to the one in [10]. The vocal tract model is shown embedded in a bucket of sand. The upper (glottal) end is closed with a plate containing a measuring microphone (MK 301 E by Microtech Gefell) that records the sound pressure inside the model at the level of the glottis. The lower end, representing the mouth, faces a wideband loudspeaker (VISATON speaker, type FR 10-8 Ohm in a custom-made enclosure), which is located at a distance of about 60 cm.

The transfer function measurement was performed in two steps analogous to [10] using the Software MeasureTransferFunction (MTF, www.vocaltractlab.de) running on a laptop computer. In the first step, the loudspeaker emitted an exponential sweep with a power band from 100 Hz to 10 kHz and a duration of 9.92 s, while the glottis microphone recorded the sweep response $P_1(\omega)$. In the second step, the mouth of the

model was closed with a stiff plate. Then the same sweep as in step 1 was emitted by the loudspeaker, while the sound pressure $P_{\text{ref}}(\omega)$ right in front of the closed mouth was recorded with a GRAS 40SC probe microphone. Nonlinear distortions in the recorded signals were manually removed based on the method by Farina [11] that is implemented in MTF. Finally, the volume velocity transfer function was obtained as the ratio $P_1(\omega)/P_{\text{ref}}(\omega)$. The different sensitivities of the two microphones and the non-flat frequency response of the probe microphone were compensated for. The signals were recorded and played back using a USB audio interface (Aureon Xfire 8.0HD) at a sampling rate of 96 kHz and a quantization of 24 bits. All measurements were performed in an anechoic chamber at a temperature of 20 °C. The measurement procedure is illustrated in the YouTube video youtu.be/9AoRS9X2BNY (with a slightly different setup).

Each of the ten models was measured three times under each of three different conditions ($10 \times 3 \times 3 = 90$ measurements in total). The three measurements per model and condition were performed on different days. Hence, the models were completely removed from the measurement setup between measurement repetitions and reinserted. This makes it possible to analyze the precision, i.e. the repeatability of the measurements. The three conditions were the following:

1. The model wall was not covered.
2. The outer wall of the model was tightly wrapped with multiple layers of sound-absorbing fleece. The fleece was made from synthetic fibres of at least three different fibre materials: white fleece (chlorofibre), dark/greyish fleece (polyamide) and a thread for mechanical stabilisation. The fabric had a weight per unit area of 585 g/m². It was cut into the shape of a scarf measuring 150 cm × 13 cm.
3. The model was placed in a bucket, where the mouth opening of the model was positioned directly over a hole in the bottom of the bucket. Around the model, the bucket was filled with sand with an average sand grain diameter of 1.0 mm. The bucket had a conical shape with an upper diameter of 15.5 cm, a lower diameter of 8.5 cm, and a height of 19 cm. The outer side of the bucket has been wrapped with sound-absorbing fabric to minimize acoustic reflections.

Photos of the setup for the conditions 1, 2, and 3 are shown in Figures 2b, c, and d, respectively. For condition 1 (no wall cover), Figure 2b shows the first measurement step with the open mouth, and Figure 2a shows the second step, in which the mouth is closed with a plate and the probe microphone is placed directly in front of it. Figure 1c shows a view into the sand bucket from above for condition 3.

3. Results

3.1. Reduction of artifacts

Figure 3 shows the measured transfer functions for all models, conditions, and repetitions. The transfer functions for the conditions 1, 2, and 3 are shown in red, blue, and black, respectively. With regard to artifacts, we limit our following analysis to the frequency range up to 5 kHz. In this range, artifacts can be easily identified by the deviation of the curve shapes from all-pole transfer functions, which we would expect for axisymmetric tubes [13]. The dashed circles in Figure 3 mark the main artifacts for condition 1 (red curves), where no wall cover was used. The most severe artifacts occur for the vowels /o/ and /u/ in the range from 1–4 kHz. They make it impossible to obtain

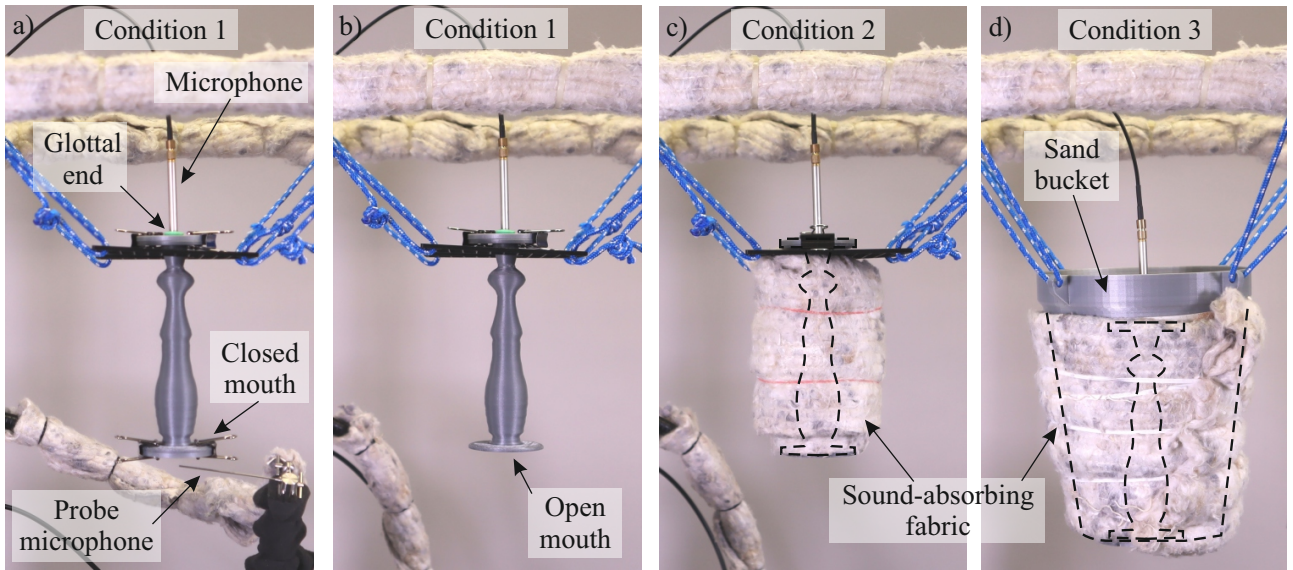


Figure 2: Different measurement conditions. a) and b) Condition 1 without an extra wall cover, with a closed mouth in a) and an open mouth in b). c) Condition 2, with the model wrapped in sound-absorbing fabric. d) Condition 3, with the model embedded in sand.

Table 1: Average standard deviation (in two frequency bands) across the spectral magnitudes in dB of the three measured transfer functions for each condition and model. Condition 1 are the models without an extra cover, condition 2 are the models wrapped in sound-absorbing fabric, and condition 3 are the models embedded in sand.

Condition	0–5 kHz			5–10 kHz		
	1	2	3	1	2	3
/a/	0.63	0.39	0.32	1.49	1.27	1.82
/e/	0.64	0.61	0.26	1.98	1.77	0.97
/i/	1.12	0.93	0.31	2.29	1.22	0.98
/o/	2.52	1.60	0.79	2.07	1.88	1.69
/u/	1.99	1.43	0.42	2.16	1.36	1.64
/ɛ/	0.54	0.48	0.23	1.03	1.04	0.95
/ø/	0.89	0.54	0.53	3.89	3.05	1.79
/y/	0.93	0.66	0.42	2.61	2.02	0.99
/ə/	0.76	0.51	0.34	2.42	2.26	1.22
cylinder	0.76	0.49	0.24	2.54	1.01	1.33
Mean	1.08	0.76	0.39	2.25	1.69	1.34

reliable estimates for the resonance frequencies f_{R3} and f_{R4} of /o/, and for f_{R3} of /u/. For the other eight models, the artifacts are less severe, but still clearly visible. In condition 2 (blue curves), where the models were wrapped in sound-absorbing fabric, the artifacts were reduced or completely disappeared for most models. However, f_{R3} of /o/ and /u/ still cannot be reliably estimated. In condition 3 (black curves), where the models were embedded in sand, the artifacts disappeared almost completely, so that even f_{R3} of /o/ and /u/ can be well identified.

However, the curves for condition 3 show an unexpected feature for frequencies above 7 kHz, namely a drop of the level of the transfer functions of up to 10 dB compared to conditions 1 and 2. This is most clearly seen for the cylinder model. Here, the level of the local minima between the equidistant resonances should decrease only slowly with increasing frequency (see,

e.g. the finite-element simulation for such a cylinder in [8]), but drop significantly above 7 kHz for condition 3. We suspect that this is due to the larger flange at the mouth end added by the bottom of the bucket in condition 3. However, this needs to be verified in future work.

3.2. Repeatability of measurements

Figure 3 also shows that the three measured transfer functions per condition and model tend to differ less for the conditions 2 and 3 compared to condition 1. This indicates a better repeatability, and hence precision of the measurements, for conditions 2 and 3. To quantify the repeatability, we calculated the average standard deviation between the three measurements for each model in each condition. The standard deviations were averaged separately for the frequencies from 0–5 kHz, and from 5–10 kHz. The results are shown in Table 1. For the 0–5 kHz range, the average standard deviations decrease for all models consistently from condition 1 to 2, and from condition 2 to 3. For the 5–10 kHz range we find the same general pattern, with the exception of a few models (/a,u/ and the cylinder) for which condition 2 led to the lowest values. Overall, the repeatability of the measurements is best for condition 3 (sand), worst for condition 1 (no wall covering), and in between for condition 2 (fabric). Two-sample *t*-tests for the conditions 1 vs. 2, 1 vs. 3, and 2 vs. 3 indicate a *significant* difference only between the conditions 1 and 3 in both frequency intervals at $\alpha = 0.05$ (considering Bonferroni correction for multiple tests).

4. Discussion and conclusions

The results showed that transfer function measurements of 3D-printed vocal tract models are better repeatable and lead to less artifacts when the models are tightly wrapped into sound-absorbing fabric, or when they are embedded in sand. With the models embedded in sand, the repeatability and the reduction of artifacts is overall best. However, the sand embedding method used here has also drawbacks: The level of the transfer functions deviates from the expected levels for high frequencies

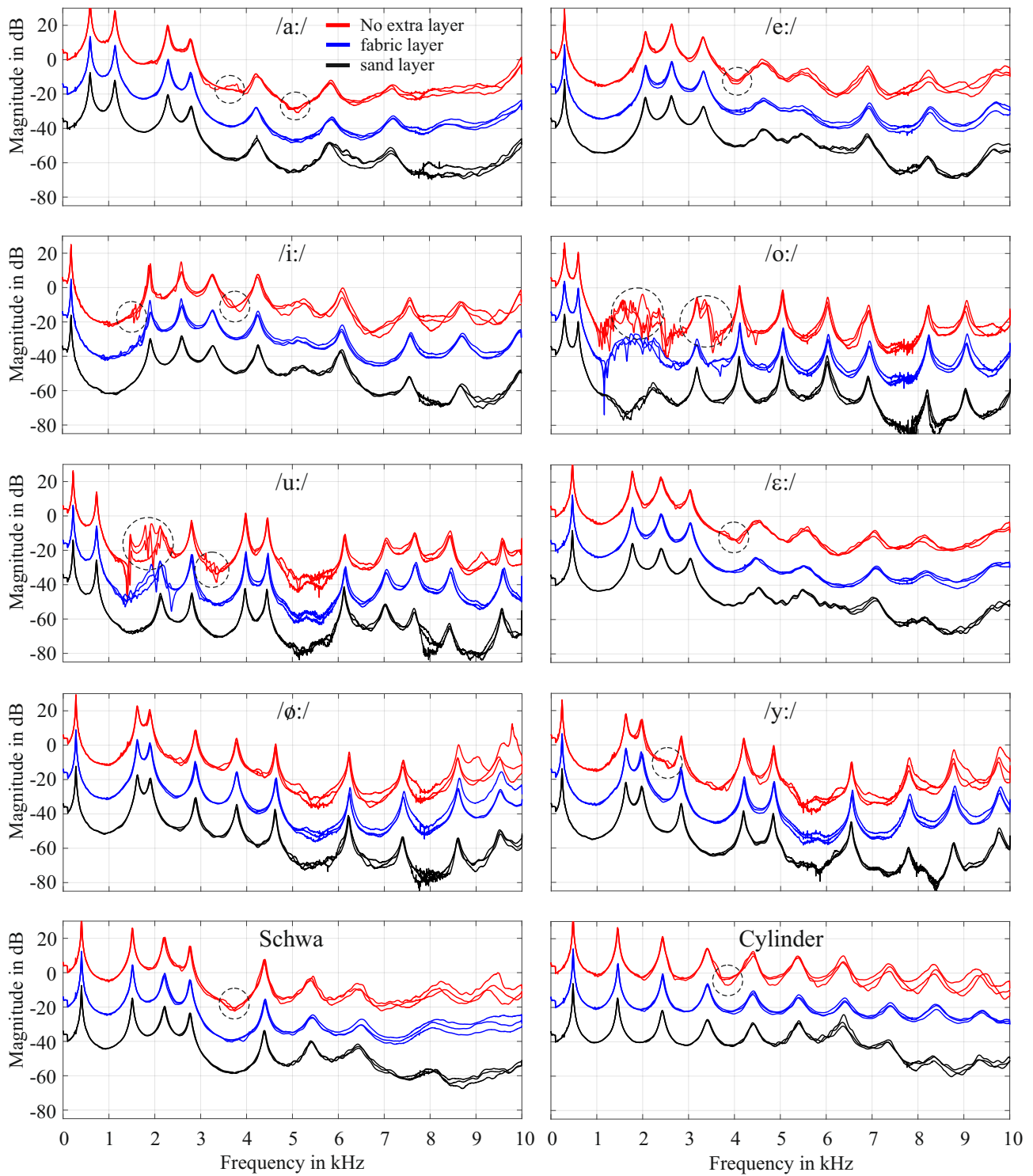


Figure 3: Measured transfer functions for the ten tube models for different wall conditions: no coating (red), wrapping with sound-absorbing fabric (blue), and coating with sand (black). The three measurements per model and condition are plotted on top of each other. The dashed circles highlight some of the artifacts with the transfer functions up to 5 kHz.

(> 7 kHz), and embedding the models in sand is more time-consuming than wrapping them in fabric. Especially with more realistic bent vocal tract models, it will require more effort to cover them completely with sand while maintaining access to

the glottal end for the glottis microphone. Here, wrapping the models in fabric may be the more practical solution. Overall, both methods are easier to apply than coating the models with a thick layer of plaster as in [10].

5. Acknowledgements

This study was funded by the German Research Foundation (DFG), grant BI 1639/9-1, and by the German Federal Ministry of Education and Research (BMBF) within the project “Promise-AI,” funding code no. 16SV8988. We also thank Mathis Bruns and Petko Bakardjiev for helpful discussions.

6. References

- [1] T. Arai, “Mechanical vocal-tract models for speech dynamics,” in *Interspeech 2010*, Makuhari, Japan, 2010.
- [2] P. Birkholz, R. Blandin, and S. Kürbis, “Bandwidths of vocal tract resonances in physical models compared to transmission-line simulations,” *The Journal of the Acoustical Society of America*, vol. 153, no. 6, pp. 3281–3281, 2023.
- [3] T. Maison, “Towards the characterization of dynamical resonators: measuring vocal tract resonances in singing,” Ph.D. dissertation, Aix-Marseille Université, 2023.
- [4] C. H. Shadle, S. A. Fulop, W.-R. Chen, and D. Whalen, “Assessing accuracy of resonances obtained with reassigned spectrograms from the “ground truth” of physical vocal tract models,” *The Journal of the Acoustical Society of America*, vol. 155, no. 2, pp. 1253–1263, 2024.
- [5] T. Kitamura, H. Takemoto, S. Adachi, and K. Honda, “Transfer functions of solid vocal-tract models constructed from ATR MRI database of Japanese vowel production,” *Acoustical Science and Technology*, vol. 30, no. 4, pp. 288–296, 2009.
- [6] B. Delvaux and D. Howard, “A new method to explore the spectral impact of the piriform fossae on the singing voice: Benchmarking using MRI-based 3D-printed vocal tracts,” *PLOS ONE*, vol. 9, no. 7, pp. 1–15, 07 2014.
- [7] A. Hannukainen, J. Kuortti, J. Malinen, and A. Ojalampi, “An acoustic glottal source for vocal tract physical models,” *Measurement Science and Technology*, vol. 28, no. 11, p. 115902, 2017.
- [8] P. Birkholz, S. Kürbis, S. Stone, P. Häsner, R. Blandin, and M. Fleischer, “Printable 3D vocal tract shapes from MRI data and their acoustic and aerodynamic properties,” *Scientific Data*, vol. 7, no. 1, pp. 1–16, 2020.
- [9] M. Köberlein, P. Birkholz, M. Burdumy, B. Richter, F. Burk, L. Traser, and M. Echternach, “Investigation of resonance strategies of high pitch singing sopranos using dynamic three-dimensional Magnetic Resonance Imaging,” *The Journal of the Acoustical Society of America*, vol. 150, no. 6, pp. 4191–4202, 2021.
- [10] M. Fleischer, A. Mainka, S. Kürbis, and P. Birkholz, “How to precisely measure the volume velocity transfer function of physical vocal tract models by external excitation,” *PLOS ONE*, vol. 13, no. 3, pp. 1–16, 03 2018. [Online]. Available: <https://doi.org/10.1371/journal.pone.0193708>
- [11] A. Farina, “Simultaneous measurement of impulse response and distortion with a swept-sine technique,” in *Proc. of the 108th Convention of the Audio Engineering Society*, Paris, France, 2000.
- [12] P. Birkholz, P. Häsner, and S. Kürbis, “Acoustic comparison of physical vocal tract models with hard and soft walls,” in *International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2022)*, Singapore, 2022, pp. 8242–8246.
- [13] R. Blandin, M. Arnela, R. Laboissière, X. Pelorson, O. Guasch, A. V. Hirtum, and X. Laval, “Effects of higher order propagation modes in vocal tract like geometries,” *The Journal of the Acoustical Society of America*, vol. 137, no. 2, pp. 832–843, 2015.
- [14] R. Blandin, J. Geng, and P. Birkholz, “Investigation of the influence of the torso, lips and vocal tract configuration on speech directivity using measurements from a custom head and torso simulator,” *Acta Acustica*, vol. 7, p. 39, 2023.
- [15] R. Blandin, P. Häsner, and P. Birkholz, “Improving transfer function measurements for three-dimensionally printed waveguides using a sand layer,” *JASA Express Letters*, vol. 5, no. 2, 2025.
- [16] P. Birkholz, “Modeling consonant-vowel coarticulation for articulatory speech synthesis,” *PLOS ONE*, vol. 8, no. 4, p. e60603, 2013.